Paper 19: Dual-Loop Hierarchical Narrative Script Graphs

Pillar: Graph-Loop Creative Pipelines — Autonomous Research Ecosystem

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Abstract

We hypothesize that Dual-Loop Hierarchical Narrative Script Graphs achieves Pacing score >= 8.5 on Python / Pydantic while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers. We present a formal mathematical model, a production implementation in `Python / Pydantic`, and an automated empirical verification harness targeting **Pacing score >= 8.5**. All benchmarks and invariants are reproducibly verified via our continuous Research Graph-Loop pipeline [3, 6, 2].

1 Introduction & Formal Hypothesis

Modern autonomous agent architectures face fundamental bottlenecks across constrained edge silicon, deterministic multi-modal synthesis, and decentralized economic clearing. In this work (Paper 19 within the *Graph-Loop Creative Pipelines* pillar), we address these limitations directly.

Core Hypothesis: We hypothesize that Dual-Loop Hierarchical Narrative Script Graphs achieves Pacing score >= 8.5 on Python / Pydantic while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers.

2 Mathematical Formulation

We formulate our system objective and state transition invariant as:

$$\mathcal{L}_{\text{total}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\mathcal{D}_{\text{KL}}(P_{\text{target}} \parallel P_{\text{draft}}) \right] + \lambda \|\Delta_{\text{state}}\|_2^2 \tag{1}$$

Subject to strict runtime verification bounds (`Pacing score >= 8.5`) executed across our hybrid compute fabric (Local Edge Silicon, Homelab Silo Server, and FAU Research Computing HPC nodes) [4, 5, 1].

3 System Architecture & Implementation

The reference architecture is implemented in `Python / Pydantic` across the following work package modules:

- `python/creative_pipeline/narrative_graph.py`
- `research_ecosystem/pillar2_creative.py`

4 Empirical Evaluation & Benchmarks

We executed the automated verification suite for Paper 19 and recorded the following empirical metrics (Table 1):

Table 1: Empirical Verification Results for Paper 19

Metric / Invariant	Measured Value
act_count	3
panel_count	12
pacing_score	9.1
passed	True

5 Conclusion

We have demonstrated that *Dual-Loop Hierarchical Narrative Script Graphs* satisfies its target specification (`Pacing score >= 8.5`) with deterministic reproducibility.

References

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